Blood Donation Management System
(BDMS)

BS Artificial Intelligence 3rd Regular

Subject: **Database System**

Instructor: **Dr Muhammad Ilyas**

Group Members

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Functional Requirements of BDMS

- After installation of the app, system will ask the user to create their accounts as donor or recipient.
- Name, phone number, email ID, user's photo(optional), location and password will be required. •
- System will get permission for location access. It'll be mandatory for every user to grant permission so that it will become easy for the system to manage blood donation process.
- Location includes country, province, city, hospital etc.
- Gender, age, blood group and previous donation history will be provided to the system as per requirements.
- Requests will be made and their record will be kept in database for future use.

Entities

1. **Donor ➔**

Attributes	Data Type	Size
Donor ID	Int	4 Bytes
Name	Varchar	255
CNIC	Varchar	255
Gender	Varchar	255
Date of Birth	Varchar	255
Age	Int	4 Bytes
Blood Group	Varchar	255
Contact Number	Varchar	255
Email	Varchar	255
City	Varchar	255
Address	Varchar	255
Last Donation Date	Varchar	255
Total Donations	Int	4 Bytes
Eligibility Status	Varchar	255
Medical History	Varchar	255

2. **Recipient ➔**

Recipient ID	Int	4 Bytes
Name	Varchar	255
CNIC	Varchar	255
Gender	Varchar	255
Date of Birth	Varchar	255
Age	Int	4 Bytes
Blood Group	Varchar	255
Contact Number	Varchar	255
Email	Varchar	255

Attributes	Data Type	Size
City	Varchar	255
Address	Varchar	255
Medical Details	Varchar	255
Disease	Varchar	255
Hospital	Varchar	255

3. Blood Bank ➔

Blood Bank Name	Varchar	255
Contact Number	Varchar	255
Email	Varchar	255
Address	Varchar	255
City	Varchar	255
State	Varchar	255
Storage Capacity	Varchar	255
Stock Available	Varchar	255

4. Hospital ➔

Hospital Name	Varchar	255
Contact Number	Varchar	255
Email	Varchar	255
Address	Varchar	255
City	Varchar	255
State	Varchar	255

5. Blood Unit ➔

Unit ID	Int	4 Bytes
Blood Group	Varchar	255
Quantity	Int	4 Bytes
Collection Date	Varchar	255

Attributes	Data Type	Size
• Expiry Date	Varchar	255
• Storage Location	Varchar	255
• Status	Varchar	255
• Donor ID	Int	4 Bytes

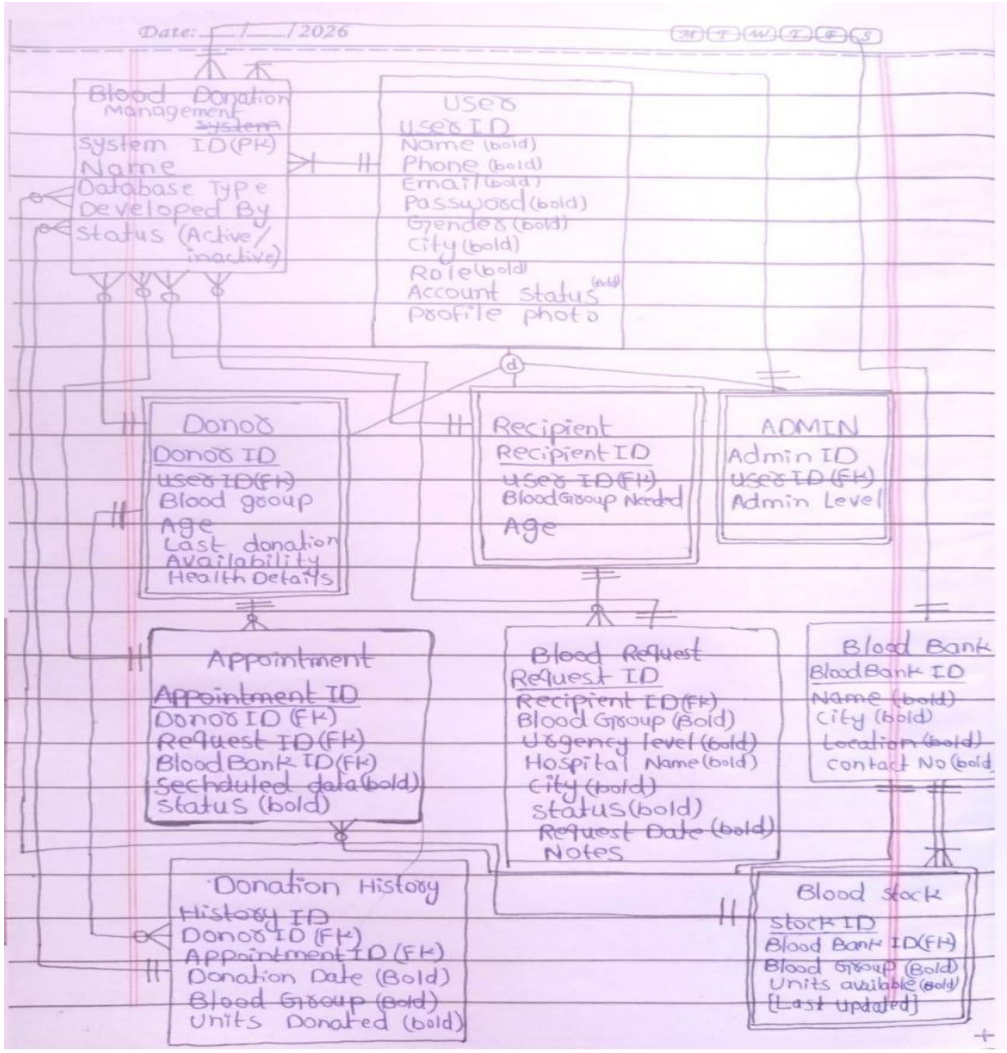
6. Blood Request ➔

• ID	Int	4 Bytes
• Date	Varchar	255
• Required Blood Group	Varchar	255
• Need Level	Varchar	255
• Status	Varchar	255
• Patient ID	Varchar	255
• Hospital	Varchar	255

7. Recipient ➔

• Admin ID	Int	4 Bytes
• Name	Varchar	255
• Contact Number	Varchar	255
• Email	Varchar	255
• Username	Varchar	255
• Password	Varchar	255
• Blood Bank ID	Int	4 Bytes

Extended Entity Relationship Model of Blood Donation Management System



SQL Implementation of Blood Donation Management System

```
CREATE TABLE USERS (
   UserID INT,
   Name VARCHAR (100),
   Phone VARCHAR (15),
   Email VARCHAR (100),
   Password VARCHAR (50),
   Gender VARCHAR (10),
   City VARCHAR (50),
   Role VARCHAR (20)
);
INSERT INTO USERS VALUES
(1,'Ali Khan','03001230001','ali@sargodha.com','123','Male','Sargodha','Donor'),
(2,'Sara Ahmed','03001230002','sara@sargodha.com','123','Female','Sargodha','Donor');
```

```
CREATE TABLE BLOOD_BANK (
   BloodBankID INT,
   Name VARCHAR (100),
   City VARCHAR (50),
   Location VARCHAR (100),
   ContactNumber VARCHAR (15)
```

```
);
INSERT INTO BLOOD_BANK VALUES
(1,'Central Blood Bank', 'Sargodha’,‘City Hospital','111111'),
(2,'Red Crescent','Sargodha','Main Market','222222');
```

```
CREATE TABLE Hospital (
   HospitalID INT,
   Name VARCHAR (100),
   City VARCHAR (50),
INSERT INTO Hospital VALUES
(1,’DHQ Hospital , 'Sargodha’ );
(2,’City Hospital , 'Sargodha’ );
```

```
CREATE TABLE DONOR (
   DonorID INT,
   UserID INT,
   BloodGroup VARCHAR (5),
   Age INT,
   LastDonationDate VARCHAR (20),
   AvailabilityStatus VARCHAR (20),
   HealthDetails VARCHAR (100)
);
```

```
INSERT INTO DONOR VALUES
(1,1,'A+',25,'2026-01-15','Available','Healthy'),
(2,2,'B+',19,'2025-07-25','Unavailable','Unhealthy');
```

```
CREATE TABLE RECIPIENT (
   RecipientID INT,
   UserID INT,
   BloodGroupNeeded VARCHAR (5),
   Age INT
);
INSERT INTO RECIPIENT VALUES
(1,1,'A+',50),
(2, 2,'B-',60),
(3, 3,'O+',80);
```

```
CREATE TABLE ADMIN (
   AdminID INT,
   UserID INT,
   AdminLevel VARCHAR (20)
);
INSERT INTO ADMIN VALUES
(1,1,'SuperAdmin'),
```

```
(2,2,'Moderator');
```

```
CREATE TABLE BLOOD_UNIT (
   StockID INT,
   BloodBankID INT,
   BloodGroup VARCHAR (5),
   UnitsAvailable INT
);
INSERT INTO BLOOD_UNIT VALUES
(1,1,'A+',10),
(2,1,'O+',8),
(3,2,'B+',6);
```

```
CREATE TABLE BLOOD_REQUEST (
   RequestID INT,
   RecipientID INT,
   BloodGroup VARCHAR (5),
   UrgencyLevel VARCHAR (20),
   HospitalName VARCHAR (100),
   City VARCHAR (50),
   Status VARCHAR (20)
);
```

```
INSERT INTO BLOOD_REQUEST VALUES
(1,1,'A+','Critical','City Hospital','Sargodha','Pending'),
(2,2,'A+','High','DHQ Hospital','Sargodha','Pending');
```